

LCTC: Lightweight Convolutional Thresholding Sparse Coding Network Prior for Compressive Hyperspectral Imaging

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Abstract—Compressive spectral imaging has garnered significant attention for its ability to effectively enhance the captured spatial and spectral information. Predominant methods, based on compressive sensing, typically formulate the imaging task as a constrained optimization problem and rely on hand-crafted priors to model the sparsity of spectral images. However, these approaches often suffer from suboptimal performance due to the inherent difficulty of identifying an appropriate transform space where spectral images exhibit sparsity. To overcome this limitation, we propose a novel convolutional sparse coding-inspired untrained network prior for fast and adaptive identification of the sparse transform domain and compressible signal. Specifically, a Lightweight Convolutional Thresholding sparse Coding (LCTC) network is designed as the sparse transform domain, with its inputs interpreted as sparse coefficients. Crucially, both the transform domain and its coefficients are solved in a self-supervised learning manner. Furthermore, we demonstrate that LCTC prior can be seamlessly incorporated into the iterative optimization algorithm as a Plug-and-Play (PnP) regularization. Both the LCTC and PnP-LCTC exhibit superior performance compared to previous methods. Experiments under various scenarios validate the effectiveness and efficiency of our approach.

Index Terms—Hyperspectral imaging, computational imaging, inverse problem, compressive sensing.

I. INTRODUCTION

HYPERSPECTRAL imaging techniques capture tens to hundreds of narrow spectral bands across a wide range of electromagnetic wavelengths from observed scenes. The high spectral resolution of hyperspectral images (HSIs) enables detailed characterization of object properties and

composition, facilitating in-depth analysis in diverse fields, including remote sensing [1], medical diagnostics [2], and face recognition [3]. Hyperspectral imaging systems typically rely on spatial or spectral scanning mechanisms to acquire HSI data [4]. While these methods are well-established, challenges such as slow data acquisition speeds and bulk hardware pose significant obstacles to practical applications.

To mitigate these limitations, snapshot spectral imaging techniques, particularly Compressive Spectral Imaging (CSI) or Spectral Compressive Imaging (SCI), have emerged as alternatives by leveraging computational imaging methods [5], [6]. CSI techniques can capture spatial and spectral information in a single or few exposures, offering advantages such as high acquisition efficiency and reduced data volume, which are ideal for dynamic scene analysis applications. Generally, CSI follows the ‘encoding-decoding’ paradigm. High-dimensional HSIs data is first encoded by the synergism of dispersive prisms with binary coded apertures [7], color-coded masks [8], pattern-specific masks [9], or spatial light modulators [10]. Subsequently, algorithms are applied to decode (reconstruct) spectral images from the encoded measurements. However, reconstructing HSIs from compressed measurements is inherently an under-determined problem, making it challenging to achieve high-fidelity reconstructions.

Researchers commonly rely on compressive sensing [11] to tackle the ill-posed imaging problem. The fundamental idea is to exploit the compressibility or sparsity of HSIs in specific transform domains. This characteristic of sparsity is explicitly incorporated into optimization algorithms. Considerable efforts have been made to find an appropriate transform domain and formulate an optimization problem to solve the corresponding coefficients. For instance, the Generalized Alternating Projection with Total Variation (GAP-TV) algorithm [12] assumes that HSIs are sparse in the gradient domain and adopts an alternating projection method to iteratively estimate coefficients. Decompress Spectral Compressive Imaging (DeSCI) [13] considers that HSIs are sparse in the low-rank subspace and utilizes the Alternating Direction Method of Multipliers (ADMM) [14] optimization algorithm. The Convolutional Sparse Coding (CSC) [15], [16] is introduced in [17] for spectral imaging, leveraging the idea that HSIs can be sparsely represented using convolutional kernels and their coefficients. While these methods are effective when the selected transform domain aligns well with the data,

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choosing a universal and adaptive transform domain remains a fundamental challenge due to the diverse characteristics of natural scenes.

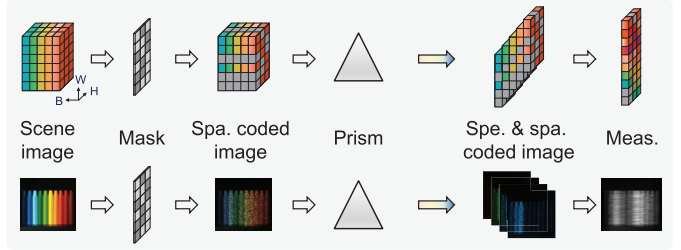
Instead of manually selecting sparse transform domains, deep neural networks have been explored for directly mapping encoded measurements to spectral images [18], [19], [20]. Despite achieving promising results, these methods rely on large-scale training datasets to ensure robust generalization across different hardware configurations. Alternatively, self-supervised learning methods, especially Untrained Neural Network Prior (UNNP) based methods [21] such as Deep Image Prior (DIP) [22] and Deep Hyperspectral Prior (DHP) [23], have been developed for generating spectral images without pre-training on large-scale datasets. Moreover, Meng et al. [24] and Chen et al. [25] have extended DIP into traditional iterative optimization algorithms for improving performance.

Nevertheless, existing UNNP-based methods (or their variations) face three key limitations: i) lack of interpretability: previous approaches neglect the physical meaning of untrained neural networks, leading to reduced interpretability compared to traditional compressive sensing methods; ii) high computational cost: excessive parameterization and suboptimal network designs result in slow reconstruction speeds, preventing practical applicability; iii) sub-par performance: current UNNP-based methods struggle to achieve state-of-the-art reconstruction quality and exhibit insufficient robustness to noise in real-world scenarios.

In this paper, we propose a convolutional sparse coding (CSC)-inspired untrained network prior to address the above challenges. First, a novel Lightweight Convolutional Thresholding sparse Coding (LCTC) untrained network is designed. We consider the LCTC as a sparse transformation domain, similar to those used in compressive sensing-based methods. Correspondingly, the input of LCTC is taken as sparse coefficients. Second, the LCTC network, with only 0.24MB parameters and 4.7G floating-point operations (FLOPs), is efficiently implemented. Compared with existing UNNP-based methods, the reconstruction time is significantly reduced. More importantly, the LCTC prior, optimized in a self-supervised learning manner, demonstrates leading performance. In addition, we present that the LCTC prior can be solo utilized or incorporated into optimization algorithm frameworks, *e.g.*, Alternating Direction Method of Multipliers (ADMM) [14], as a Plug-and-Play (PnP) regularization. In conclusion, the contributions of this paper are summarized as follows:

- We propose a novel model-inspired data-driven prior for compressive spectral imaging, which assumes the untrained neural network as the sparsity transform domain and its input as the compressible signal.
- The designed Lightweight Convolutional Thresholding sparse Coding (LCTC) prior is effective and efficient for implementation. Moreover, the reconstruction performance and robustness are significantly improved.
- We present that our proposed LCTC, as a PnP prior, can be incorporated with the iterative optimization methods. Both the solo utilization of LCTC and PnP-LCTC demonstrate superior performance and efficiency.

(a) The schematic diagram of SD-CASSI



(b) The schematic diagram of DD-CASSI

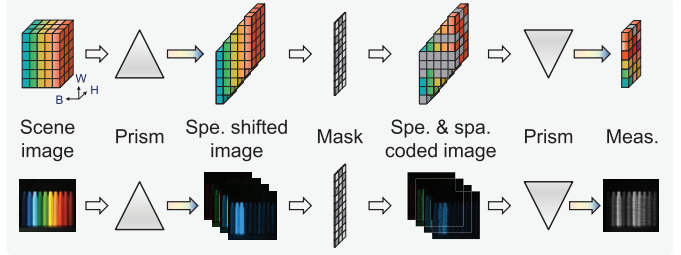


Fig. 1. (a) The schematic diagram of SD-CASSI; (b) the schematic diagram of DD-CASSI ('Spa.', 'Spe.', and 'Meas.' represent spatial, spectral, and measurements, respectively).

In summary, LCTC combines the interpretability of model-based approaches with the learning ability of neural networks. Benefiting from the proposed network architecture and sparsity optimization method, LCTC paves the way for applying self-supervised learning method to CSI. Comprehensive experiments in compressive spectral imaging, including snapshot spectral imaging (Fig. 1) and fast line-scan hyperspectral imaging (Appendix III in the supplemental material), validate the superiority of our method.

II. RELATED WORKS

A. Compressive Spectral Imaging Techniques

Compressive spectral imaging systems generally follow an 'encoder-decoder' framework. This part provides a detailed introduction to the 'encoder' component. One of the pioneering compressive sensing-based spectral imaging systems is the Dual-Disperser Coded Aperture Snapshot Spectral Imaging (DD-CASSI) introduced by Gehm et al. [7]. This system employs a binary mask for spatial modulation and two dispersive prisms for spectral modulation, operating in the 520–590 nm spectral range. Building on DD-CASSI, Wagadarikar et al. [26] proposed the Single Disperser (SD) CASSI, simplifying the system while enhancing spectral resolution. Later, Arce et al. [8] extended the CASSI by using a color-coded aperture in place of the traditional binary-coded apertures for improving reconstruction quality. Wu et al. [27] designed a multi-shot snapshot spectral imaging system utilizing a Digital Micromirror Device (DMD) instead of a static physical mask. This approach allows for dynamic modulation, capturing multiple frames with varying DMD patterns for enhanced performance. More recently, Saragadam et al. [10] proposed a scene-adaptive spectral compressive imaging system by utilizing panchromatic side information. In this paper, we evaluate the proposed algorithm on the SD-CASSI and DD-CASSI systems. However, the approach

is readily adaptable to other computational spectral imaging systems.

B. Iterative Optimization-Based Reconstruction

The iterative optimization-based spectral image reconstruction algorithms leverage the concept of compressive sensing, namely reconstructing the HSIs by solving under-determined inverse problems. They typically formulate the restoration task as a constrained optimization problem, incorporating prior knowledge about the sparsity or compressibility of the spectral data. The concept of total variation (TV) regularization was introduced by Yuan [12] as a way to maintain the spatial smoothness of HSIs. Liu et al. [13] considered the patch-wise low-rank prior, incorporating a weighted nuclear norm for low-rank regularization. Yang et al. [28] explored the Gaussian Mixture Model (GMM) of patch-based images to improve restoration results. In recent works, the deep network has been utilized as a plug-and-play regularization in the optimization framework [29]. Meng et al. [24] extended the concept of Deep Image Prior (DIP) to spectral compressive imaging. He et al. [30] introduced a self-attention mechanism-based deep image prior for increasing the interpretability of DIP in the HSIs reconstruction problem. Chen et al. [31] formulated the snapshot compressive imaging task as a low-rank and deep image prior regularized optimization problem. Chen et al. [25] further proposed a method that combines Low-Rank subspace representation and spatial-attention-based Self-supervised Deep Network (LRSDN) for reconstructing spectral images.

C. Supervised Learning-Based Reconstruction

Supervised deep learning-based hyperspectral images (HSIs) restoration methods aim to learn an end-to-end mapping function that can reconstruct HSIs directly from incomplete measurements. Miao et al. [18] proposed λ -net, a pioneering supervised learning-based CASSI approach. To further enhance the imaging quality, Meng et al. [32] developed the spatial-spectral self-attention network (TSA-Net). Their method leverages self-attention mechanisms to capture complex spatial-spectral dependencies. Zhang et al. [19] trained a neural network that first learns the information from external datasets and then finetuned the network to progressively generate spectral images with spatial-spectral constraints. Huang et al. [33] introduced a learnable Gaussian Scale Mixture (GSM) prior, which is captured by deep neural networks. This approach effectively models the statistical properties of HSIs for improved reconstruction. Wang et al. [34] designed a deep neural network architecture that unrolls a non-local self-similarity-based optimization algorithm. In recent works, Cai et al. [20], [35] proposed Transformer architecture-based hyperspectral image reconstruction networks. These networks leverage the Transformer to capture long-range dependencies in HSIs. Existing supervised learning methods demonstrate the power of deep neural networks in accurately reconstructing HSIs and have the potential to outperform traditional model-based approaches. However, they also have disadvantages, such as dependency on large-scale labeled training data, limited adaptability, and lack of interpretability.

III. PRELIMINARY

A. Mathematical Model of CSI

In this section, we present the forward observation model for compressive spectral imaging, using the Single Disperser Coded Aperture Snapshot Spectral Imaging (SD-CASSI) system as an example. While the optical configuration of other computational spectral imaging systems, such as the Dual Disperser Coded Aperture Snapshot Spectral Imaging (DD-CASSI), differ from SD-CASSI, their forward observation models can be similarly formulated. Furthermore, LCTC is not restricted to compressive spectral imaging. It is extensible to a wide range of spectral imaging inverse problems, including hyperspectral denoising (experiments can be seen in Appendix III).

The SD-CASSI system typically consists of an objective lens, a physical binary mask (coded aperture), a dispersive prism, relay lenses, and a grayscale detector. The imaging process involves spatial modulation of the scene by the coded aperture, followed by spectral dispersion introduced by the prism. The modulated and dispersed light is then integrated and captured by the grayscale detector. A schematic of the forward observation process is shown in Fig. 1.

Mathematically, let $X(i, j, b)$ denote the spectral irradiance of the scene at the coded aperture plane, where (i, j) indicate the spatial coordinates ($1 \leq i \leq H, 1 \leq j \leq W$) and b is the index of spectral dimension ($1 \leq b \leq B$). The coded aperture is described by a binary function $\mathbf{M}(i, j)$, where $\mathbf{M}(i, j) = 1$ allows light to pass through and $\mathbf{M}(i, j) = 0$ blocks it. The mask modulates the incoming light spatially. The dispersive element (e.g. a prism or diffraction grating) induces a wavelength-dependent spatial shift. For a wavelength corresponding to spectral index b , the light is shifted by $D(b)$ in the j -direction on the sensor. This shift is expressed by

$$D(b) = k_d(b - b_R), \quad (1)$$

where k_d represents the dispersion coefficient and b_R is a reference wavelength, typically the starting wavelength. The detector records the intensity after modulation and dispersion. The measurement $Y(u, v)$ at the focal plane of the sensor is given by

$$Y(u, v) = \iiint X(i, j, b) \mathbf{M}(i, j) \delta(u - i) \delta(v - j - D(b)) di dj db, \quad (2)$$

where (u, v) are the sensor coordinates, and $\delta(\cdot)$ denotes the Dirac delta function.

In practical applications, the continuous scene radiance and measurements are discretized. Throughout this paper, matrices or high-order data are represented by bold uppercase letters, e.g., $\mathbf{X}$, vectors by bold lowercase letters, e.g., $\mathbf{x}$, and scalar values by regular letters, such as H or i .

We define $\mathbf{M} \in \mathbb{R}^{H \times W}$, $\mathbf{X} \in \mathbb{R}^{H \times W \times B}$, $\mathbf{X}' \in \mathbb{R}^{H \times W \times B}$, $\mathbf{X}'' \in \mathbb{R}^{H \times (W + D(B)) \times B}$, and $\mathbf{Y} \in \mathbb{R}^{H \times (W + D(B))}$ as the discretized mask, desired hyperspectral images, spatially coded images, spectrally-spatially coded images, and measurements, respectively. The $\mathbf{X}'$ is obtained by element-wise multiplication of $\mathbf{X}$ and $\mathbf{M}$

$$\mathbf{X}'(:, :, b) = \mathbf{X}(:, :, b) \odot \mathbf{M}, \quad (3)$$

where $X(:, :, b)$ represents the b -th spectral band of X , and $\odot$ denotes the Hadamard product. The tensor X'' can be modeled by spatial shifting X' with a shifted distance $D(b)$

$$X''(i, j + D(b), b) = X'(i, j, b). \quad (4)$$

Accordingly, the measurement captured by the detector can be expressed as

$$Y(u, v) = \sum_{b=1}^B T(u, v, b) X''(u, v, b) + E(u, v), \quad (5)$$

where $T(u, v, :) \in \mathbb{R}^B$ represents the spectral response function of the pixel (u, v) and $E \in \mathbb{R}^{H \times (W + D(B))}$ denotes the imaging system noise. For simplicity, we assume the spectral response function is uniform across all pixels.

Moreover, we use the $\text{vec}(\cdot)$ operator (column-wise vectorization) to obtain $\mathbf{y} = \text{vec}(Y) \in \mathbb{R}^{N_y}$, $\mathbf{x} = \text{vec}(X) \in \mathbb{R}^{N_x}$, and $\mathbf{e} = \text{vec}(E) \in \mathbb{R}^{N_y}$, respectively. Here, $N_y = H(W + k_d(B - b_R))$, $N_x = HWB$, and k_d is in units of spatial shift per unit spectral band. Hence, Eq. (5) can be rewritten as

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{e}, \quad (6)$$

where $\mathbf{A} \in \mathbb{R}^{N_y \times N_x}$ is the sensing matrix, which is defined as

$$\mathbf{A} = [\mathbf{M}_1, \dots, \mathbf{M}_b, \dots, \mathbf{M}_B], \quad (7)$$

where $\mathbf{M}_b$ denotes the block diagonal of vectorized mask $\mathbf{M}$ after applying the spectral shift corresponding to the b -th spectral band.

B. Reconstruction Problem and Challenges of CSI

The reconstruction problem is solved by maximizing the posterior probability $p(\mathbf{x}|\mathbf{y})$. According to Bayes' theorem, it can be expressed as

$$p(\mathbf{x}|\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{x}) \cdot p(\mathbf{x})}{p(\mathbf{y})}. \quad (8)$$

Taking the logarithm of both sides, the problem of estimating the desired hyperspectral images $\mathbf{x}$ becomes

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x}} \log p(\mathbf{y}|\mathbf{x}) + \log p(\mathbf{x}), \quad (9)$$

where $p(\mathbf{y}|\mathbf{x})$ represents the likelihood term for quantifying the discrepancy, and $p(\mathbf{x})$ represents the prior term, serving as a regularization constraint.

Most existing methods assume that the discrepancy follows a Gaussian distribution and spectral images are sparse in a certain transform domain. Consequently, the optimization problem is formulated as

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \lambda g(\mathbf{x}), \quad (10)$$

where $\|\cdot\|_2^2$ denotes the square ℓ_2 norm, $g(\mathbf{x})$ is the sparse regularization, and λ is a balancing factor.

Following the concept of compressive sensing [36], $g(\mathbf{x})$ can be formulated as

$$\min_{(\Phi, \theta)} \|\theta\|_1, \quad \text{subject to } \mathbf{x} = \Phi\theta \quad (11)$$

where Φ signifies the sparsity transformation matrix, θ indicates the compressible signal and the desired hyperspectral images are obtained as $\mathbf{x} = \Phi\theta$. Substituting this into the optimization problem yields

$$\begin{aligned} \hat{\mathbf{x}} &= \arg \min_{\theta} \frac{1}{2} \|\mathbf{y} - \mathbf{A}\Phi\theta\|_2^2 + \lambda \|\theta\|_1, \\ \text{subject to } \mathbf{x} &= \Phi\theta. \end{aligned} \quad (12)$$

Existing methods aim to solve the problem described in Eq. (12) by selecting an appropriate transform domain Φ as a regularization. Fig. 2a illustrates the framework of model-based methods. Since the system model $\mathbf{A}$ is fixed when scenes $\mathbf{x}$ change, it is vital to design an appropriate and generalized transformation matrix Φ . On the other hand, supervised learning-based methods involve training an end-to-end neural network that directly maps the measurement $\mathbf{y}$ to the collected reference spectral image $\mathbf{x}$. As depicted in Fig. 2b, parameters of the neural network are trained by minimizing the loss function

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2, \quad (13)$$

where $\hat{\mathbf{x}}$ represents the predicted spectral images. However, supervised learning methods heavily rely on large-scale training datasets. Moreover, the performance of these methods tends to degrade when applied to systems with different setups.

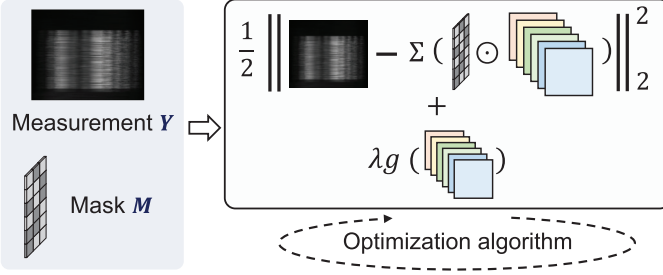
IV. THE PROPOSED METHOD

This paper proposes a novel untrained neural network prior that combines the strengths of model-driven and data-driven approaches. Unlike traditional methods that rely on predefined sparsity transforms, our prior adaptively finds the optimal sparse transform domain according to the captured scene. Specifically, a Lightweight Convolutional Thresholding sparse Coding (LCTC) untrained network prior is designed as the sparsity transformation matrix Φ , with its input serving as the compressible signal θ , as illustrated in Fig. 3a. Thus, the desired spectral images are obtained as $\mathbf{x} = \Phi(\theta)$. In the following sections, we first present the core component of our prior, *i.e.*, the proposed Convolutional Thresholding sparse Coding (CTC) mechanism. Next, we provide details of the implementation of LCTC prior for the CSI and then demonstrate its integration as a Plug-and-Play (PnP) prior within iterative optimization algorithms.

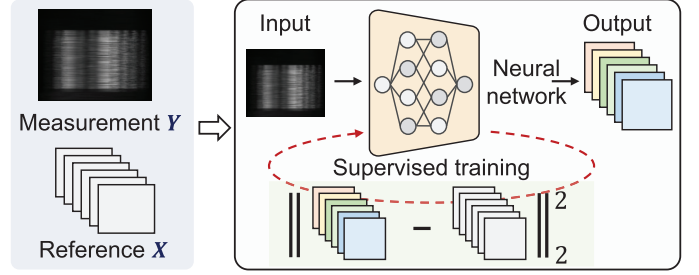
A. Convolutional Thresholding Sparse Coding Mechanism

The Convolutional Thresholding sparse Coding (CTC) module is specifically designed for implementing the untrained neural network prior with addressing two key challenges: i) lack of interpretability in untrained network priors-based methods; ii) high computational cost. Inspired by the theory of Convolutional Sparse Coding (CSC), CTC comprises two components, namely convolutional dictionaries ($\mathbf{D}_{\text{Conv}}$ as highlighted with a blue background in Fig. 3b) and thresholding sparse coefficients (highlighted with a gray background in Fig. 3b). Specifically, the convolutional dictionaries include three convolutional layers ($\text{Conv}(\cdot)$), two activation functions ($\text{Act}(\cdot)$). The first convolutional layer uses a kernel size of

(a) Model-based method



(b) Deep learning-based method



(c) The proposed untrained neural network prior

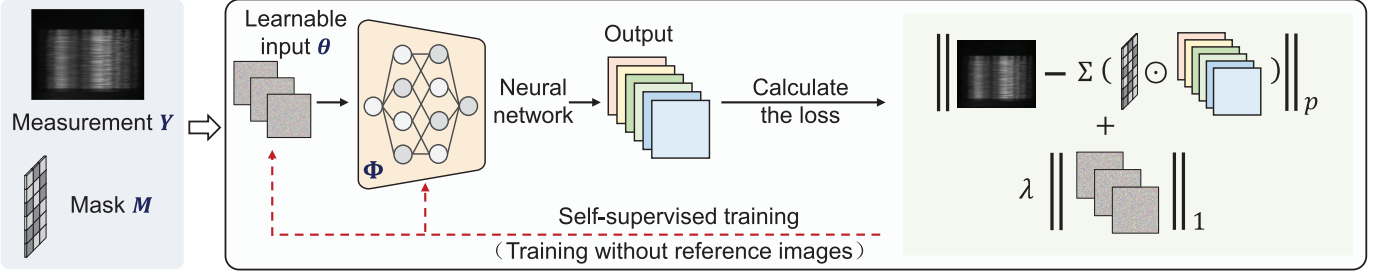
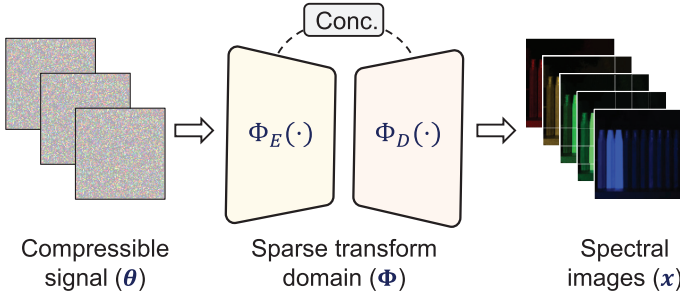


Fig. 2. Existing compressive spectral imaging restoration methods can be categorized as (a) model-based and (b) supervised learning-based methods. (c) The proposed algorithm combines the advantage of both the model-based and supervised learning-based methods, which achieves high-performance reconstruction results without training on large-scale datasets.

(a) The framework of LCTC prior



(b) CTC mechanism

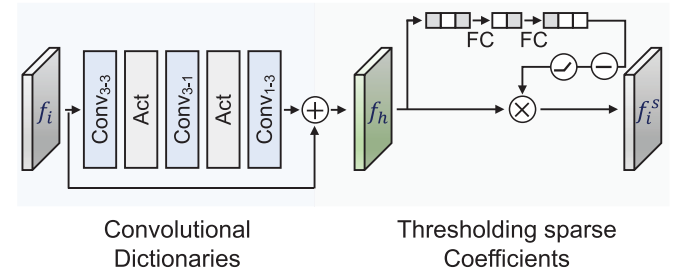
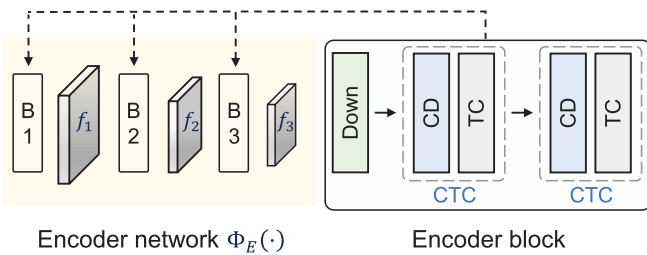
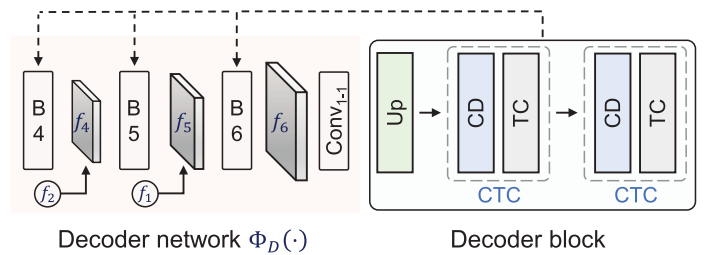
(c) $\Phi_E(\cdot)$ network architecture(d) $\Phi_D(\cdot)$ network architecture

Fig. 3. (a) The framework of the LCTC prior, where ‘Conc.’ denotes the skip connections; (b) The convolutional thresholding sparse coding module architecture; (c) the network architecture of $\Phi_E(\cdot)$; (d) the network architecture of $\Phi_D(\cdot)$.

(3, 3), denoted as Conv_{3-3} in Fig. 3. The last two convolutional layers are set with a kernel size of (3, 1) and (1, 3), denoted as Conv_{3-1} and Conv_{1-3} respectively, to reduce parameters without sacrificing performance. Each layer outputs 36 channels ($c = 36$), and the effect of c is analyzed in the supplemental material II-A. Residual connection [37] (denoted as $\oplus$ in Fig. 3b) further improves optimization and training efficiency. Overall, given convolutional dictionaries ($\mathbf{D}_{\text{Conv}}$) and sparse

coefficients (f_i), representations are obtained as

$$f_h = f_i \oplus \text{Conv}(\text{Act}(\text{Conv}(\text{Act}(\text{Conv}(f_i))))). \quad (14)$$

More importantly, this paper proposes a thresholding strategy to get the sparse feature maps f_i , effectively. Considering that the channel dimension corresponds to the spectral dimension and spectral bands are always redundant, we reduce the trivial spectral information of feature maps via a thresholding

significance weight $s_h \in \mathbb{R}^c$, where c is the number of channels. Specifically, the weight s_h can be obtained by

$$\begin{aligned}\tilde{s}_h &= \text{Sigmoid}(\text{FC}(\text{Act}(\text{FC}(\text{GP}(f_h))))), \\ s_h &= S_t(\tilde{s}_h),\end{aligned}\quad (15)$$

where $\text{FC}(\cdot)$ is the Fully-Connected layer. The first FC layer maps features with dimensions c to c/r , where r is the reduction ratio. The second FC layer restores the dimension from c/r to c . Additionally, $\text{GP}(\cdot)$ represents the global pooling and $\text{Soft}_t(\cdot)$ denotes the soft-shrinkage operation

$$\text{Soft}_t(\beta) = \begin{cases} \beta - t, & \text{if } \beta - t \geq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

The soft-shrinkage operation is innovatively implemented by subtraction and rectified linear unit activation function operation ($\text{Soft}_t(\beta) \equiv \text{ReLU}(\beta - t)$), as shown in Fig. 3b. In conclusion, given the input f_h , we leverage the thresholding strategy to generate sparse coefficients f_i^s by

$$f_i^s = f_h \otimes S_h, \quad (17)$$

where $\otimes$ denotes element-wise multiplication and S_h represents the spatially repeated significance weights. It should be noted that the output f_i^s are treated as the sparse coefficients for the next convolutional dictionaries. Thus, by stacking convolutional thresholding sparse coding modules, we can consider the network as a sparse transform domain Φ , which progressively transforms input sparse coefficients to the desired spectral images $\hat{x} = \Phi * \theta$.

B. LCTC Prior for CSI

With the proposed lightweight CTC module, we now introduce the overall architecture of the LCTC untrained network and the application to compressive spectral imaging. Generally, the LCTC network (*i.e.*, Φ) adopts a U-shape architecture, consisting of an encoder network $\Phi_E(\cdot)$ for reducing the size of features maps and a decoder network $\Phi_D(\cdot)$ for reconstruction, as depicted in Fig. 3a. The encoder network $\Phi_E(\cdot)$, shown in Fig. 3c, consists of 3 blocks (denoted as B1-B3). Each block includes a downsampling operator ($\text{Down}(\cdot)$) and two designed Convolutional Thresholding sparse Coding mechanisms ($\text{CTC}(\cdot)$). The output feature maps of the $(k+1)$ -th block (f_{k+1}) are obtained through

$$f_{k+1} = \text{CTC}(\text{CTC}(\text{Down}(f_k))). \quad (18)$$

The shape of the initial input θ (or f_0) is $B^I \times H \times W$. To minimize the number of parameters, we set $B^I = 3$ (details of experiments are provided in Appendix II-A of the supplemental material).

Similarly, the decoder network $\Phi_D(\cdot)$ comprises three blocks, denoted as B4-B6. Each block includes an upsampling operator ($\text{Up}(\cdot)$) and two CTC modules. To facilitate gradient flow and preserve information across layers, skip connections are employed to bridge corresponding

features from the encoder and decoder networks. The output feature maps of each decoder block are obtained as follows

$$\begin{aligned}\tilde{f}_k &= \text{Up}(\text{Conc}(f_k, f_{6-k})), \\ f_{k+1} &= \text{CTC}(\text{CTC}(\tilde{f}_k)),\end{aligned}\quad (19)$$

where $\text{Conc}(\cdot)$ denotes the concatenation operation.

Finally, the output of decoder network $\Phi_D(\cdot)$ is convolved with a $(1, 1)$ kernel and reshaped to form the predicted spectral images

$$\hat{x} = \Phi_D(\Phi_E(\theta)) := \Phi(\theta). \quad (20)$$

In summary, the proposed LCTC untrained network is highly compact, with only 0.24MB of parameters and 4.7G floating-point operations (FLOPs). Its lightweight design enables significantly faster reconstruction speeds compared to existing untrained neural network prior-based methods.

1) *Optimization of LCTC Network*: Most previous Compressive Spectral Imaging (CSI) algorithms utilize the ℓ_2 norm (*i.e.*, least squares error) as the data fidelity term, based on the assumption that the likelihood $p(y|x)$ follows a standard Gaussian distribution

$$p(y|x) = \mathcal{N}(y|x, \sigma^2) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{\|y - Ax\|_2^2}{2\sigma^2}\right). \quad (21)$$

Maximizing the likelihood is equivalent to minimizing the least squares error. However, while suitable for Gaussian noise removal, the ℓ_2 norm often leads to over-smoothed images and is suboptimal for the complex noise characteristics in CSI.

In this paper, non-Gaussian noises are considered, which have the characteristic of sparse and are always caused by the device's hardware. Intuitively, the ℓ_0 norm, namely $\|y - Ax\|_0$, should be minimized. However, the ℓ_0 norm is non-differentiable and intractable, which increases the difficulty and instability of optimization. The ℓ_1 norm, a convex relaxation of the ℓ_0 norm, is more practical and has been shown in prior works [38], [39] to be equivalent to ℓ_0 minimization under certain conditions, such as when A is a random matrix with independent standard Gaussian entries, as is the case in CSI. Therefore, we define the loss function for the LCTC prior as

$$\mathcal{L}_{\text{LCTC}} = \|y - A\hat{x}\|_1 + \lambda \|\theta\|_1, \quad (22)$$

where λ is a balancing parameter, and $\hat{x}$ represents the reconstructed spectral images. Comparative experiments (Section V-C) demonstrate that the ℓ_1 norm significantly outperforms the ℓ_2 norm.

The parameters of the LCTC network (*i.e.*, the sparsity transform domain Φ) and the compressible signal (the input sparse coefficients θ) are optimized end-to-end. The network weights are initialized using the Kaiming initialization method [40], and the effect of initialization is discussed in Appendix II-D of the supplemental material. The parameters of θ are initialized with random Gaussian noise (see Appendix B for visualization θ during training). Optimization is performed using the Adam optimizer [41], a stochastic gradient descent algorithm. After training with E_p epochs, the final output is taken as the desired hyperspectral images ($x = \Phi^{E_p}(\theta^{E_p})$).

C. PnP-LCTC Algorithm for CSI

In this section, we demonstrate how the proposed LCTC prior can be employed as a Plug-and-Play (PnP) prior to enhance the performance of iterative optimization algorithms. Moreover, with proper configuration, the computational cost of the PnP-LCTC optimization algorithm remains competitive.

Accordingly, PnP-LCTC regularization term is defined as

$$g(\mathbf{x}) = \|\mathbf{y} - \mathbf{A}\Phi(\boldsymbol{\theta})\|_1 + \lambda\|\boldsymbol{\theta}\|_1, \quad (23)$$

where the LCTC prior can be seen as the combination of a data-driven ($\|\mathbf{y} - \mathbf{A}\Phi(\boldsymbol{\theta})\|_1$) and model-driven prior ($\|\boldsymbol{\theta}\|_1$). Incorporating this prior, the traditional optimization problem is reformulated as

$$\hat{\mathbf{x}} = \arg \min_{(\mathbf{x}, \Phi, \boldsymbol{\theta})} \frac{\lambda_1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \underbrace{\|\mathbf{y} - \mathbf{A}\Phi(\boldsymbol{\theta})\|_1 + \lambda\|\boldsymbol{\theta}\|_1}_{\text{LCTC prior}}. \quad (24)$$

where λ_1 represents a balancing factor. Furthermore, additional regularization, such as total variation [12], [42] and low-rank [13], can also be integrated into this framework. For simplicity, we adopt the ADMM [14] algorithm to solve the above problem, although PnP-LCTC can be extended to accommodate other regularization techniques.

The scaled augmented Lagrangian function for this problem is defined as

$$\mathcal{L}_\rho(\mathbf{x}, \mathbf{v}, \mathbf{u}) = \frac{\lambda_1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + g(\mathbf{v}) + \frac{\rho}{2} \|\mathbf{x} - \mathbf{v} + \mathbf{u}\|_2^2, \quad (25)$$

where $\rho \in \mathbb{R}$ is a balancing factor, $\mathbf{v} \in \mathbb{R}^{N_x}$ and $\mathbf{u} \in \mathbb{R}^{N_x}$ denote the auxiliary variable and scaled Lagrangian multiplier, respectively. The scaled augmented Lagrangian function is then solved by decomposing it into three subproblems

$$\begin{aligned} \hat{\mathbf{x}} &= \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \frac{\rho}{2\lambda_1} \|\mathbf{x} - \mathbf{v} + \mathbf{u}\|_2^2, \\ \hat{\mathbf{v}} &= \arg \min_{\mathbf{v}} g(\mathbf{v}) + \frac{\rho}{2} \|\mathbf{x} - \mathbf{v} + \mathbf{u}\|_2^2, \\ \hat{\mathbf{u}} &= \mathbf{u} + (\mathbf{x} - \mathbf{v}). \end{aligned} \quad (26)$$

The specific optimization process is presented in Appendix I of the supplemental material. For $(n+1)$ -th iteration, the variables $\mathbf{x}^{n+1}$, $\mathbf{v}^{n+1}$, and $\mathbf{u}^{n+1}$ are updated as follows

$$\begin{aligned} \mathbf{x}^{n+1} &= (\mathbf{v}^n - \mathbf{u}^n) \\ &\quad + \mathbf{A}^\top [\mathbf{y} - \mathbf{A}(\mathbf{v}^n - \mathbf{u}^n)] \oslash \left[\text{diag}(\mathbf{A}\mathbf{A}^\top) + \frac{\rho}{\lambda_1} \right], \\ \mathbf{v}^{n+1} &= \Phi^{Ep}(\boldsymbol{\theta}^{Ep}), \\ \mathbf{u}^{n+1} &= \mathbf{u}^n + (\mathbf{x}^{n+1} - \mathbf{v}^{n+1}) \end{aligned} \quad (27)$$

Both the $\mathbf{x}^{n+1}$ and $\mathbf{u}^{n+1}$ have closed-form solutions. And the solution of $\mathbf{v}^{n+1}$ is shown in Section IV-B. The entire process of the PnP-LCTC is summarized in Algorithm 1.

D. Convergence Analysis of PnP-LCTC

Here, we demonstrate that the proposed PnP-LCTC algorithm exhibits fixed-point convergence [43], [44], i.e., $(\mathbf{x}, \mathbf{v}, \mathbf{u})$ will converge to point $(\mathbf{x}^*, \mathbf{v}^*, \mathbf{u}^*)$.

Algorithm 1 PnP-LCTC Algorithm for CSI

Require: $\mathbf{A}$, $\mathbf{y}$, p , λ_1 , λ , ρ , Ep , and N

Initial Φ , $\boldsymbol{\theta}$, $\mathbf{x}$, $\mathbf{v}$, and $\mathbf{u}$

while $n \leq N$ or not converge **do**

1: Update $\mathbf{x}^{n+1}$

while $it \leq Ep$ **do**

2.1: Take a gradient descent step on

$$\nabla_{\Phi}(\mathcal{L}_{\text{LCTC}} + \frac{\rho}{2} \|\mathbf{x}^{n+1} - \Phi(\boldsymbol{\theta}) + \mathbf{u}^n\|_2^2)$$

2.2: Take a gradient descent step on

$$\nabla_{\boldsymbol{\theta}}(\mathcal{L}_{\text{LCTC}} + \frac{\rho}{2} \|\mathbf{x}^{n+1} - \Phi(\boldsymbol{\theta}) + \mathbf{u}^n\|_2^2)$$

2.3: Update Φ^{it+1} and $\boldsymbol{\theta}^{it+1}$

2.4: $it = it + 1$

end while

3: Update $\mathbf{u}^{n+1}$

4: $n = n + 1$

end while

Output: $\mathbf{v}$

Proposition 1 (Gradient Boundedness): Let $\mathbf{x} \in [0, 1]^{N_x}$ denotes the normalized spectral images and $\mathbf{A} \in \mathbb{R}^{N_y \times N_x}$ be a linear operator with eigenvalues lying within the interval $[0, 1]$. Then, the gradient of the data fidelity term $\|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2$ is bounded by α_1 . Specifically, we have

$$\begin{aligned} \|\nabla \left(\frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 \right)\|_2 &= \|\mathbf{A}^\top (\mathbf{A}\mathbf{x} - \mathbf{y})\|_2 \\ &\leq \lambda_{\max}(\mathbf{A})(\|\mathbf{x}\|_2 + \|\mathbf{y}\|_2) := \alpha_1, \end{aligned} \quad (28)$$

where $\lambda_{\max}(\mathbf{A})$ denotes the maximum eigenvalue of $\mathbf{A}$, and α_1 is a constant upper bound on the scaled gradient norm.

Proposition 2 (Boundedness of LCTC Prior): The LCTC network is a bounded function, meaning that the deviation between its output $\mathbf{v}$ and the reference approximation $\mathbf{x} + \mathbf{u}$ is bounded by α_2 :

$$\|\mathbf{v} - (\mathbf{x} + \mathbf{u})\|_2 \leq \sqrt{L_{\text{PnP-LCTC}}} := \alpha_2, \quad (29)$$

where $L_{\text{PnP-LCTC}}$ is the objective function associated with the LCTC prior within the $\mathbf{v}$ -subproblem. This deviation progressively decreases during iterations as the LCTC network is continuously optimized.

Theorem 1: The PnP-LCTC converges to $(\mathbf{x}^*, \mathbf{v}^*, \mathbf{u}^*)$, i.e.,

$$\begin{aligned} \|\mathbf{x}^n - \mathbf{x}^*\|_2 &\rightarrow 0, \\ \|\mathbf{v}^n - \mathbf{v}^*\|_2 &\rightarrow 0, \\ \|\mathbf{u}^n - \mathbf{u}^*\|_2 &\rightarrow 0, \end{aligned} \quad (30)$$

when $n \rightarrow \infty$.

Proof: We demonstrate that $(\mathbf{x}^n, \mathbf{v}^n, \mathbf{u}^n)$ is a Cauchy sequence such that

$$\begin{aligned} \|\mathbf{x}^{n+1} - \mathbf{x}^n\|_2 &= C_x^n, \\ \|\mathbf{v}^{n+1} - \mathbf{v}^n\|_2 &= C_v^n, \\ \|\mathbf{u}^{n+1} - \mathbf{u}^n\|_2 &= C_u^n, \end{aligned} \quad (31)$$

where the $C_x^n \in \mathbb{R}$, $C_v^n \in \mathbb{R}$, and $C_u^n \in \mathbb{R}$ are decreased with iteration going on, i.e., $C_x^{n+1} < C_x^n$.

Firstly, for the $\|\mathbf{v}^{n+1} - \mathbf{v}^n\|_2$, according to the first-order optimality of $\mathbf{x}$ -subproblem, we have

$$\begin{aligned} & \|\mathbf{x}^{n+1} - (\mathbf{v}^n - \mathbf{u}^n)\|_2 \\ &= \frac{\lambda_1}{\rho} \|\nabla(\|\mathbf{y} - \mathbf{A}\mathbf{x}^{n+1}\|_2^2)\|_2 \leq \frac{2\alpha_1\lambda_1}{\rho}. \end{aligned} \quad (32)$$

Then, based on the triangle inequality, the $\|\mathbf{v}^{n+1} - \mathbf{v}^n\|_2$ follows

$$\begin{aligned} & \|\mathbf{v}^{n+1} - \mathbf{v}^n\|_2 \\ & \leq \|\mathbf{v}^{n+1} - (\mathbf{x}^{n+1} + \mathbf{u}^n)\|_2 + \|(\mathbf{x}^{n+1} + \mathbf{u}^n) - \mathbf{v}^n\|_2 \\ & \leq \alpha_2 + \frac{2\alpha_1\lambda_1}{\rho}. \end{aligned} \quad (33)$$

Secondly, for the $\|\mathbf{u}^{n+1} - \mathbf{u}^n\|_2$, according to the subproblem of $\mathbf{u}$, we have

$$\|\mathbf{u}^{n+1}\|_2 = \|\mathbf{u}^n + (\mathbf{x}^{n+1} - \mathbf{v}^{n+1})\|_2 \leq \alpha_2. \quad (34)$$

Hence, $\|\mathbf{u}^{n+1} - \mathbf{u}^n\|_2$ follows

$$\|\mathbf{u}^{n+1} - \mathbf{u}^n\|_2 \leq \|\mathbf{u}^{n+1}\|_2 + \|\mathbf{u}^n\|_2 \leq 2\alpha_2. \quad (35)$$

Moreover, the subproblem of $\mathbf{u}$ also implies that $\mathbf{x}^{n+1} = \mathbf{u}^{n+1} - \mathbf{u}^n + \mathbf{v}^{n+1}$. Finally, for the $\|\mathbf{x}^{n+1} - \mathbf{x}^n\|_2$, we have

$$\begin{aligned} & \|\mathbf{x}^{n+1} - \mathbf{x}^n\|_2 \\ &= \|(\mathbf{u}^{n+1} - \mathbf{u}^n + \mathbf{v}^{n+1}) - (\mathbf{u}^n - \mathbf{u}^{n-1} + \mathbf{v}^n)\|_2 \\ & \leq \|\mathbf{u}^{n+1} - \mathbf{u}^n\|_2 + \|\mathbf{u}^n - \mathbf{u}^{n-1}\|_2 + \|\mathbf{v}^{n+1} - \mathbf{v}^n\|_2 \\ & \leq 5\alpha_2 + \frac{2\alpha_1\lambda_1}{\rho}. \end{aligned} \quad (36)$$

In summary, we present $\|\mathbf{x}^{n+1} - \mathbf{x}^n\|_2$, $\|\mathbf{v}^{n+1} - \mathbf{v}^n\|_2$, and $\|\mathbf{u}^{n+1} - \mathbf{u}^n\|_2$ are bounded by α_2 or α_2 with α_1 , which are iteratively decreased. As a result, the sequence $(\mathbf{x}^n, \mathbf{v}^n, \mathbf{u}^n)$ is a Cauchy sequence and will converge to a fixed point $(\mathbf{x}^*, \mathbf{v}^*, \mathbf{u}^*)$. $\square$

V. EXPERIMENTS

We evaluate the performance of the proposed LCTC prior and PnP-LCTC methods by comparing them with existing algorithms on various spectral image datasets. Hyperparameter settings of LCTC and PnP-LCTC are presented in Appendix II-A of the supplemental material. The experimental datasets include 10 scenes from the KAIST dataset [45], the Toy scene from the CAVE dataset [46], and the Bird scene dataset [7]. The original KAIST dataset comprises hyperspectral images with a spatial size of 2704×3376 . In our experiments, we follow the setup in [24] and utilize 10 scenes for testing, as illustrated in Fig. 4. Each scene is cropped to a resolution of 256×256 . The original data contains 31 spectral bands, ranging from 400 nm to 700 nm with a step size of 10 nm. For consistency with the real system [32], the spectral wavelength bands are cropped to range from 450 nm to 650 nm, resulting in 28 band channels. The Toy scene has a size of $512 \times 512 \times 31$, where 512 is the image spatial size and 31 represents the number of spectral bands. The Bird hyperspectral image with a shape of $1021 \times 703 \times 24$ is shown in Fig. 4. Here, 1021×703 denotes the spatial resolution and 24 is the number of spectral bands.

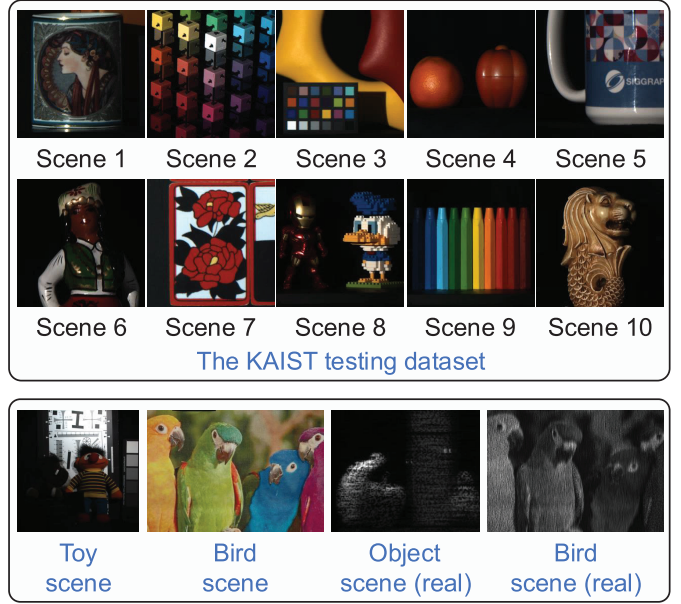


Fig. 4. The RGB reference images of simulation experimental datasets and the measurements of real-world datasets.

To evaluate the reconstruction performance, we employ two spatial image quality metrics and one spectral quality metric. Specifically, the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM) [47] are used to assess the spatial information of the reconstructed spectral images. These metrics are calculated for each spectral band image, and the average values across all bands are reported as the final results. Higher PSNR and SSIM values indicate superior reconstruction quality. Additionally, we utilize the spectral angle mapper (SAM) to evaluate the spectral accuracy of the reconstructed hyperspectral images. Lower SAM values represent better reconstruction results.

A. Simulation Experiments on SD-CASSI System

In this section, we evaluate the proposed methods through simulation experiments on the Single-Disperser Coded Aperture Snapshot Spectral Imaging (SD-CASSI) system. As depicted in Fig. 1, the SD-CASSI setup consists of a coded aperture mask and a prism. To simulate the spectral dispersion effect, a two-pixel shift is introduced between neighboring spectral bands, resulting in measurements with dimensions of 256×310 for the KAIST dataset. For the LCTC prior, the number of training epochs (Ep) is set to 4,000, striking a balance between computational efficiency and reconstruction performance. For the PnP-LCTC method, the number of outer iterations (N) is set to 7, with each iteration involving 850 epochs of LCTC network training. Importantly, the network weights are carried over between iterations rather than trained from scratch. This warm-start strategy significantly reduces the training burden per iteration (*i.e.*, $850 \ll 4,000$ as used in standalone LCTC), thereby accelerating convergence and lowering the overall computational cost. The influence of hyperparameters is detailed in Appendix II-B of the supplemental material.

TABLE I

COMPARISON OF RECONSTRUCTION PERFORMANCE ON THE 10 KAIST TESTING SCENES. THE PSNR AND SAM RESULTS ARE REPORTED (THE BEST RESULTS ARE HIGHLIGHTED IN BOLD AND THE SECOND-BEST RESULTS ARE LABELED WITH AN UNDERLINE)

Method	Scene1 PSNR	Scene2 PSNR	Scene3 PSNR	Scene4 PSNR	Scene5 PSNR	Scene6 PSNR	Scene7 PSNR	Scene8 PSNR	Scene9 PSNR	Scene10 PSNR	Average PSNR
GAP-TV	28.70	27.64	29.60	37.07	27.01	25.85	26.58	24.68	27.63	25.42	28.02
DeSCI	28.03	26.44	29.62	39.77	25.69	24.49	25.87	23.63	27.78	25.00	27.63
PnP-HSI	29.51	29.00	31.26	39.05	28.39	26.67	27.97	24.59	28.74	25.72	29.09
PnP-DIP	32.48	29.62	31.45	39.63	29.49	29.34	32.38	28.43	34.32	28.07	31.52
LRSDN	34.95	34.95	38.24	44.35	32.54	32.21	37.33	29.53	38.12	29.80	35.20
λ -Net	30.26	28.25	30.07	38.08	27.11	27.78	27.17	26.74	29.11	26.66	29.13
TSA-Net	32.80	31.64	32.96	40.01	29.94	31.62	30.79	29.83	32.21	29.73	32.15
DGSMP	33.90	32.27	33.75	41.27	30.02	33.30	31.65	31.87	32.03	32.03	33.21
MST-L	35.98	36.72	37.17	43.08	33.50	35.28	34.67	33.46	35.67	33.31	35.88
DAUHST	37.25	39.02	<u>41.04</u>	46.14	35.79	37.07	37.56	35.10	40.01	34.58	38.36
DHP	33.17	33.02	35.32	44.28	32.89	30.67	32.62	28.38	28.88	28.26	32.75
DeepTensor	34.66	32.77	36.95	46.60	32.85	31.41	36.43	29.43	35.50	29.13	34.57
LCTC	<u>37.67</u>	36.96	41.00	<u>47.79</u>	<u>36.94</u>	35.99	<u>40.52</u>	33.81	<u>41.91</u>	33.35	<u>38.59</u>
PnP-LCTC	38.30	<u>38.03</u>	42.02	48.61	37.68	<u>36.54</u>	41.19	<u>34.53</u>	42.42	<u>33.90</u>	39.32
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
GAP-TV	0.121	0.181	0.106	0.163	0.123	0.211	0.127	0.255	0.139	0.203	0.163
DeSCI	0.111	0.147	0.081	0.078	0.111	0.144	0.104	0.192	0.096	0.162	0.123
PnP-HSI	0.102	0.150	0.086	0.094	0.094	0.155	0.108	0.215	0.114	0.147	0.127
PnP-DIP	0.128	0.160	0.107	0.141	0.114	0.195	0.071	0.207	0.132	0.178	0.144
LRSDN	0.058	0.079	0.050	<u>0.067</u>	0.062	0.098	0.041	0.125	0.058	0.093	0.073
λ -Net	0.143	0.175	0.158	0.246	0.166	0.265	0.141	0.280	0.161	0.264	0.200
TSA-Net	0.088	0.103	0.073	0.087	0.068	0.980	0.076	0.114	0.076	0.097	0.088
DGSMP	0.093	0.111	0.076	0.078	0.098	0.083	0.080	0.109	0.083	0.075	0.090
MST-L	0.070	0.081	0.060	0.076	0.058	0.079	0.062	0.104	0.075	0.085	0.075
DAUHST	<u>0.052</u>	0.061	0.034	0.056	0.036	0.061	0.042	0.076	0.044	0.062	0.052
DHP	0.089	0.128	0.075	0.141	0.098	0.185	0.065	0.221	0.122	0.169	0.129
DeepTensor	0.054	0.100	0.041	0.034	0.039	0.072	0.041	0.085	0.058	0.057	0.058
LCTC	0.042	0.068	<u>0.031</u>	<u>0.032</u>	<u>0.026</u>	<u>0.043</u>	<u>0.031</u>	<u>0.058</u>	<u>0.036</u>	<u>0.039</u>	<u>0.041</u>
PnP-LCTC	0.042	<u>0.062</u>	0.029	0.027	0.024	0.042	0.033	0.056	0.033	0.034	0.038

We compare our proposed methods with a comprehensive set of state-of-the-art approaches, including two traditional iterative optimization algorithms (GAP-TV [12] and DeSCI [13]), three deep prior-based plug-and-play optimization methods (PnP-HSI [29], PnP-DIP [24], and LRSDN [25]), five supervised learning-based models (λ -Net [18], TSA-Net [32], DGSMP [33], MST-L [20], and DAUHST [35]), and two self-supervised learning-based methods (DHP [23] and DeepTensor [48]).

The results of GAP-TV [12], DeSCI [13], PnP-HSI [29], PnP-DIP [24], LRSDN [25], DHP [23], and DeepTensor [48] are obtained using our implementation of their publicly available code. The hyperparameters of each algorithm are adjusted to achieve the best results. Specifically, for GAP-TV [12], the number of iterations is set to 500 with a total variation (TV) weight of 0.1. In DeSCI [13], we adopt 40 iterations with a block matching period of 20 and a patch size of 32×32 . PnP-HSI [29] is run for 50 iterations using the same deep denoiser as in their work [29], with a TV weight of 0.1. For PnP-DIP [24], we follow the original implementation with 60 iterations and the same network configuration as

reported in [24]. LRSDN [25] is tested with 24 iterations and a rank value of 7, consistent with the settings in [25]. DHP [23] is implemented with 128 channels and 5 convolutional layers, while DeepTensor [48] adopts their DIP network architecture settings and is trained for 4,000 epochs.

For supervised learning-based models, we directly evaluate their official pre-trained models without additional fine-tuning. All experiments are conducted on a workstation equipped with an Intel Core i5-12600K CPU and an NVIDIA RTX 3080 GPU, using the PyTorch 1.10 framework.

1) *Numerical Results of the KAIST Dataset:* Table I presents the reconstruction performance results on the KAIST dataset. The solo utilization of LCTC prior achieves an average PSNR of 38.59 dB and an average SAM of 0.041. The PnP-LCTC algorithm further enhances these metrics, achieving an average PSNR of 39.32 dB, an average SSIM of 0.967, and an average SAM of 0.038. The proposed LCTC and PnP-LCTC methods outperform all compared reconstruction algorithms in terms of average PSNR and SAM. Specifically, PnP-LCTC surpasses the compared best method, DAUHST [35], with an average PSNR improvement of 0.96 dB. Notably, while the

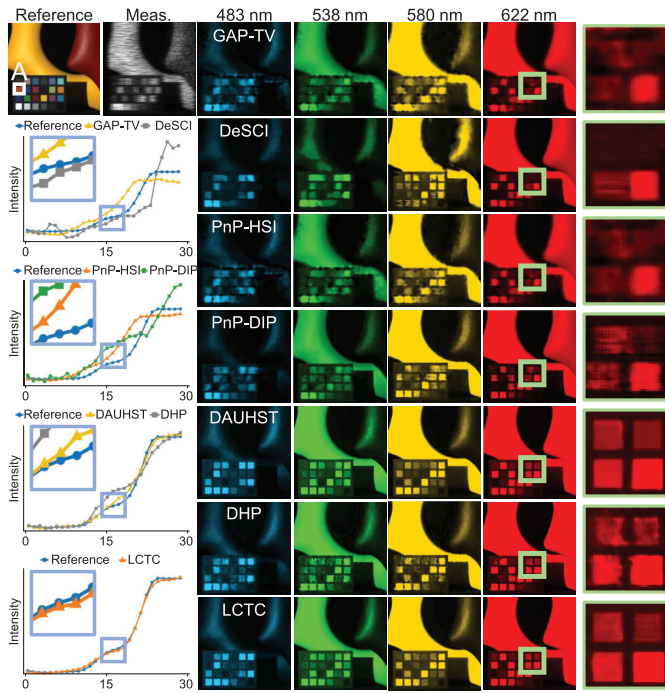


Fig. 5. Reconstructed spectral images of Scene 3 of the KAIST dataset. Additionally, one area (denoted by ‘A’) is selected for analyzing the reconstructed spectra.

DAUHST network relies on extensive training on large-scale hyperspectral datasets, our LCTC prior-based approach eliminates the need for pre-training on such datasets. Compared to the self-supervised learning (untrained neural network) based DHP [23], PnP-LCTC demonstrates a significant improvement, with a 6.57 dB increase in average PSNR and a 0.091 reduction in average SAM.

2) *Visual Results of the KAIST Dataset:* The reconstructed hyperspectral images for scene 3 of the KAIST dataset are visualized in Fig. 5. For comparison, we selected four representative spectral bands out of the 28 bands. The proposed LCTC prior consistently produces high-fidelity reconstructions with sharper details and textures compared to previous algorithms. The zoomed-in region (highlighted by the green line) further underscores the superior performance of our method, showcasing finer details and enhanced clarity. Additionally, to assess spectral reconstruction accuracy, we selected a specific point (‘A’) and compared the reconstructed spectra across different methods. The results clearly demonstrate that our algorithm achieves higher spectral fidelity and effectiveness.

B. Simulation Experiments on DD-CASSI System

In this section, we evaluate the proposed methods for the Dual-Disperser CASSI (DD-CASSI) system. The DD-CASSI setup includes a physical mask and two prisms. The dimensions of the DD-CASSI measurements of the KAIST dataset, Toy scene, and Bird scene are 256×256 , 512×512 , and 1021×703 , respectively. Since the compared supervised learning-based methods are trained based on the SD-CASSI system, our comparisons focus on iterative optimization-based and self-supervised learning-based

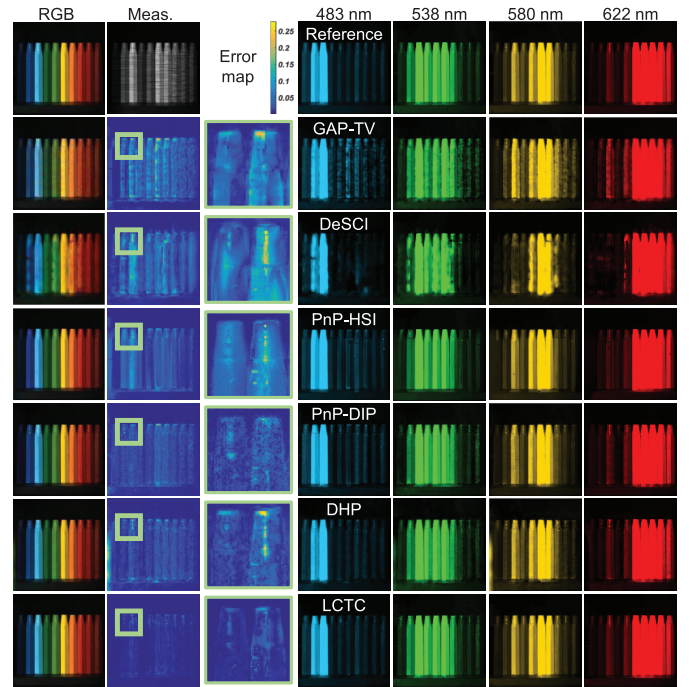


Fig. 6. Reconstructed spectral images of Scene 9 from the KAIST dataset. Moreover, the pseudo-color with an error map is illustrated (‘Meas.’ denotes measurement).

algorithms, including GAP-TV [12], DeSCI [13], PnP-HSI [29], PnP-DIP [24], LRSN [25], DHP [23], and DeepTensor [48].

1) *Numerical Results of the KAIST Dataset:* Table II presents the reconstruction performance of the 10 scenes from the KAIST dataset based on the DD-CASSI encoding system. Our LCTC model, trained for 2,000 epochs, achieves an average PSNR of 39.58 dB, an average SSIM of 0.976, and an average SAM of 0.041. The PnP-LCTC algorithm, with 5 iterations and 800 training epochs per iteration, outperforms all other methods, achieving an average PSNR of 41.29 dB, an average SSIM of 0.983, and an average SAM of 0.043. Compared to the self-supervised DHP [23], PnP-LCTC shows significant improvements of 12.34 dB in PSNR, 0.104 in SSIM, and a reduction of 0.132 in SAM. Furthermore, PnP-LCTC outperforms the deep image prior-based LRSN [25] by 2.98 dB in PSNR, 0.017 in SSIM, and 0.046 in SAM. Even with the solo utilization of LCTC, the proposed model demonstrates superior reconstruction performance compared to all previous methods.

2) *Visual Results of the KAIST Dataset:* The reconstruction results for Scene 9 of the KAIST dataset using the DD-CASSI system are visualized in Fig. 6. We selected four spectral bands for comparison. The spectral images reconstructed by the LCTC prior exhibit significantly clearer details than those produced by other methods. In addition, pseudo-color images reconstructed by each method are shown on the left side of Fig. 6. An accompanying error map, calculated as the absolute difference between the reconstructed pseudo-color images and the ground truth RGB images, highlights the superior accuracy of our method, with noticeably smaller errors compared to others.

TABLE II
COMPARISON OF RECONSTRUCTION RESULTS ON 10 SCENES OF THE KAIST DATASET. THE IMAGING SYSTEM IS THE DD-CASSI. THE PSNR, SSIM, AND SAM RESULTS ARE REPORTED

Method		Scene1	Scene2	Scene3	Scene4	Scene5	Scene6	Scene7	Scene8	Scene9	Scene10	Average
GAP-TV	PSNR	30.61	21.72	23.70	35.66	28.18	29.01	23.23	27.29	28.03	24.99	27.24
	SSIM	0.898	0.728	0.862	0.929	0.923	0.904	0.766	0.863	0.880	0.832	0.859
	SAM	0.129	0.225	0.089	0.132	0.087	0.155	0.112	0.187	0.121	0.158	0.140
DeSCI	PSNR	27.26	20.15	22.31	35.16	24.19	25.48	23.74	24.24	25.81	26.35	25.47
	SSIM	0.839	0.649	0.772	0.904	0.788	0.832	0.769	0.804	0.757	0.796	0.791
	SAM	0.181	0.246	0.149	0.156	0.229	0.237	0.110	0.269	0.185	0.242	0.200
PnP-HSI	PSNR	31.66	22.31	23.72	39.27	29.34	30.71	22.08	27.74	29.55	28.24	28.46
	SSIM	0.932	0.800	0.885	0.965	0.947	0.939	0.743	0.897	0.933	0.915	0.896
	SAM	0.098	0.160	0.077	0.114	0.078	0.125	0.066	0.164	0.089	0.129	0.110
PnP-DIP	PSNR	34.67	28.04	27.62	33.77	33.44	32.30	34.37	28.61	35.55	35.18	32.35
	SSIM	0.927	0.894	0.874	0.897	0.918	0.921	0.921	0.877	0.892	0.949	0.907
	SAM	0.110	0.253	0.119	0.209	0.126	0.196	0.088	0.258	0.134	0.182	0.168
LRSDN	PSNR	38.97	37.46	40.64	43.86	36.74	35.98	39.58	33.65	39.40	36.77	38.31
	SSIM	0.971	0.955	0.970	0.979	0.974	0.961	0.976	0.950	0.957	0.969	0.966
	SAM	0.073	0.121	0.045	0.090	0.046	0.135	0.042	0.118	0.099	0.125	0.089
DHP	PSNR	32.81	21.32	23.31	37.51	29.46	31.09	27.10	27.39	29.39	30.12	28.95
	SSIM	0.906	0.771	0.867	0.949	0.928	0.901	0.838	0.842	0.901	0.886	0.879
	SAM	0.146	0.174	0.102	0.164	0.113	0.236	0.121	0.303	0.136	0.259	0.175
DeepTensor	PSNR	37.10	36.11	39.84	45.55	37.79	37.27	40.46	35.11	38.55	36.29	38.41
	SSIM	0.975	0.951	0.969	0.984	0.975	0.977	0.978	0.962	0.958	0.974	0.970
	SAM	0.055	0.125	0.048	0.053	0.042	0.063	0.042	0.085	0.070	0.068	0.065
LCTC	PSNR	39.95	37.72	38.78	46.32	38.17	38.05	41.62	35.21	41.72	38.29	39.58
	SSIM	0.978	0.963	0.969	0.988	0.979	0.980	0.982	0.964	0.976	0.983	0.976
	SAM	0.052	0.092	0.043	0.039	0.036	0.050	0.037	0.072	0.047	0.046	0.051
PnP-LCTC	PSNR	41.65	40.21	41.88	46.77	40.74	39.34	43.34	36.35	43.34	39.25	41.29
	SSIM	0.984	0.976	0.978	0.991	0.988	0.985	0.986	0.971	0.982	0.986	0.983
	SAM	0.043	0.075	0.032	0.029	0.024	0.045	0.033	0.068	0.042	0.041	0.043

TABLE III
COMPARISON RECONSTRUCTION PERFORMANCE ON THE TOY AND BIRD SCENE DATASET

Method	Toy	Bird	Average
	PSNR/SAM	PSNR/SAM	PSNR/SAM
GAP-TV	25.26/0.163	35.11/0.042	30.19/0.103
DeSCI	26.62/0.112	37.98/0.033	32.30/0.073
PnP-HSI	24.98/0.188	26.26/0.069	25.62/0.129
PnP-DIP	28.07/0.220	26.83/0.137	27.45/0.179
LRSDN	39.21/0.087	42.17/0.024	40.69/0.056
DHP	30.92/0.176	37.68/0.033	34.30/0.105
DeepTensor	34.90/0.078	40.74/0.028	37.82/0.053
LCTC	39.32/0.064	42.91/0.021	41.12/0.043
PnP-LCTC	39.85/0.064	44.01/0.019	41.93/0.042

3) *Numerical Results of the Toy and Bird Scene:* Table III summarizes the reconstruction performance in terms of PSNR and SAM for the Toy and Bird scenes encoded by the DD-CASSI system. On the Toy scene, PnP-LCTC achieves a PSNR of 39.85 dB and a SAM of 0.064. On the Bird scene, PnP-LCTC achieves a PSNR of 44.01 dB and a SAM of 0.019, surpassing all compared algorithms. Compared to LRSDN [25], PnP-LCTC achieves a 1.24 dB average PSNR improvement across the two scenes and reduces the average

SAM by 0.014. Against the best self-supervised learning-based method, DHP [23], PnP-LCTC demonstrates a improvement of 7.63 dB in average PSNR and 0.063 in average SAM.

4) *Visual Results of the Toy and Bird Scene:* We present the reconstructed 20th spectral band image of the Toy scene in the top part of Fig. 7 and the 20th band from the Bird scene in the bottom part. The results clearly demonstrate that the spectral images reconstructed by our proposed LCTC and PnP-LCTC algorithms exhibit superior clarity and detail, such as the text in the zoomed-in areas, compared to other methods. Additionally, we present the PSNR and SSIM values for the selected spectral band image reconstructed by each algorithm. The highest PSNR and SSIM values achieved by our methods underscore their effectiveness. For instance, the proposed PnP-LCTC method achieves a PSNR of 41.45 dB and an SSIM of 0.993 on the Toy scene, significantly outperforming all competing algorithms.

C. Discussion

1) *Algorithm Running Time and Network Size:* The size and complexity of network architectures are critical factors in the practical application of deep learning-based, especially untrained neural networks-based, methods. In this section, we compare the network parameters and floating-point operations (FLOPs) of various deep learning-based algorithms alongside

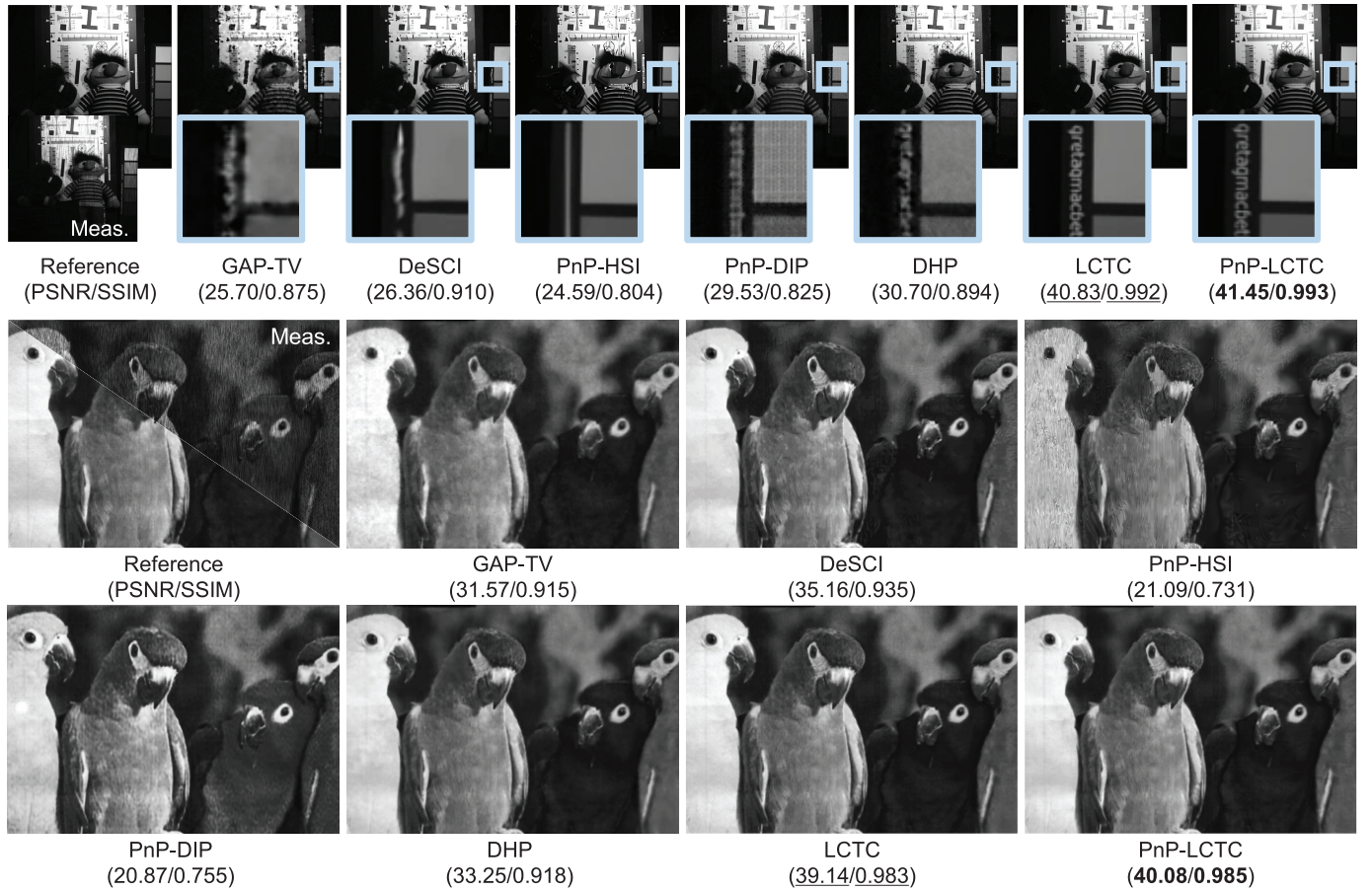


Fig. 7. Visual comparison of reconstruction results with PSNR and SSIM results on the 20th band of the Toy scene (top) and the 20th band of the Bird scene (bottom).

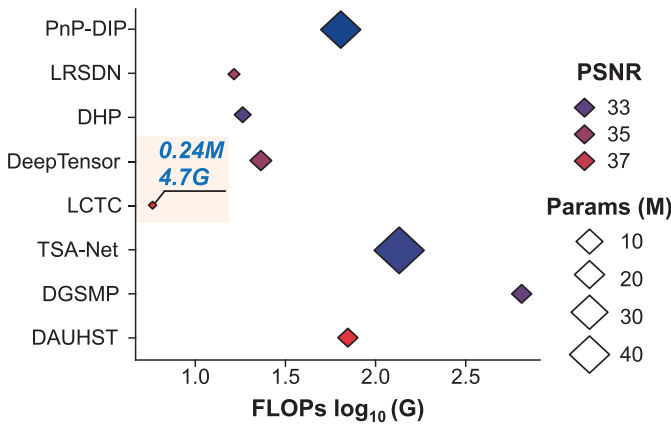


Fig. 8. Comparison network parameters and floating-point operations with the average PSNR on the KAIST dataset of deep learning-based methods.

their average PSNR performance on the KAIST dataset using the SD-CASSI system. This analysis highlights the trade-off between model complexity and reconstruction performance. Fig. 8 presents the comparison results, where the proposed LCTC network architecture demonstrates exceptional efficiency with a compact network size of just 0.24 MB and the lowest FLOPs of 4.7G, while maintaining superior PSNR values compared to other state-of-the-art methods.

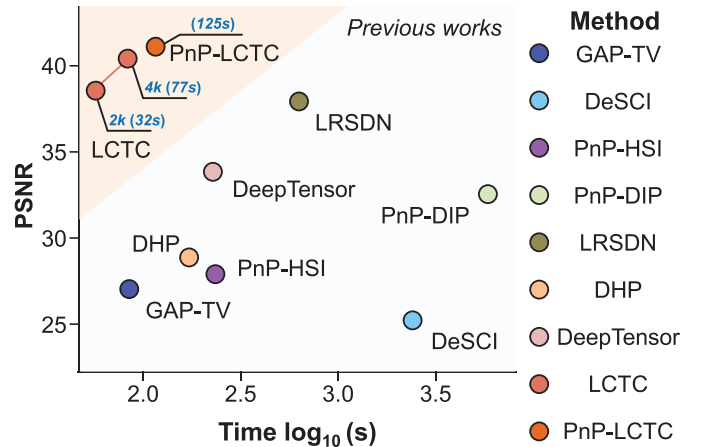


Fig. 9. Comparison algorithm running time with the average PSNR on the KAIST dataset of model-based and self-supervised learning-based methods.

In addition to network complexity, computational efficiency is another key metric for evaluating algorithm performance. We assess the running time of model-based and self-supervised learning-based methods, comparing them with their average PSNR performance on the KAIST dataset using the DD-CASSI system. It is important to note that algorithm running times can vary depending on the specific hardware configurations. For this study, GAP-TV [12] and DeSCI [13] are

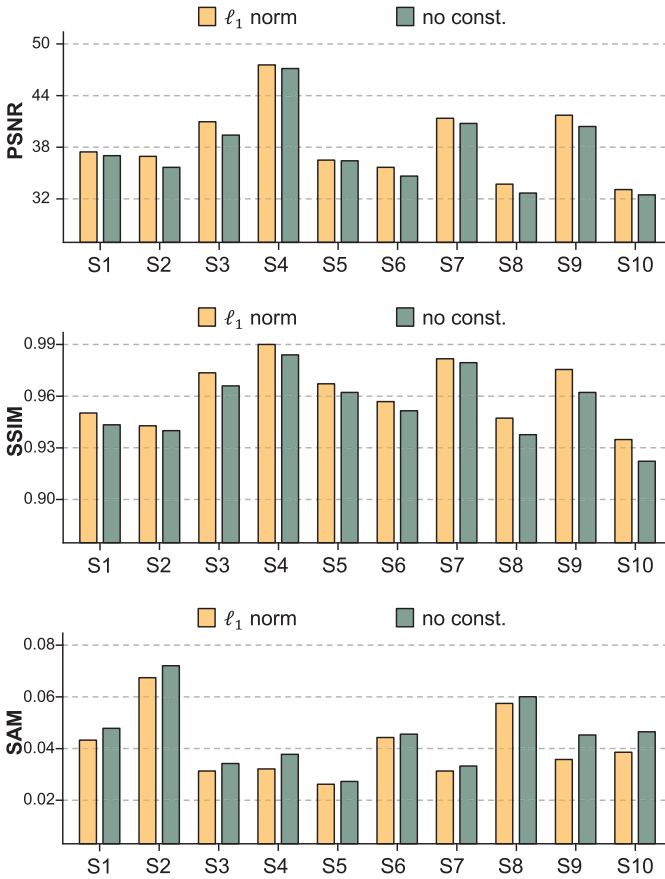


Fig. 10. The comparison of PSNR, SSIM, and SAM between the LCTC prior with ℓ_1 norm constraint and without the sparsity constraint (no const.) for network input (S1 to S10 denotes Scene 1 to Scene 10).

implemented based on their code on a CPU, while all other algorithms utilize the same GPU for a fair comparison. The results, illustrated in Fig. 9, reveal significant differences in computational efficiency. The standalone LCTC prior, trained for 2,000 epochs, achieves a remarkable running time of 32 seconds, outperforming PnP-DIP [24] by a factor of nearly 200 and DHP [23] by a factor of 6. Additionally, the PnP-LCTC algorithm delivers not only the fastest running time among GPU-based methods but also achieves the highest PSNR performance, demonstrating its superior balance of speed and reconstruction quality.

2) *Effect of the Sparsity Constraint for Network Input:* As discussed in Section IV-B, the compressive signal (network input θ) of the proposed LCTC prior is optimized under a sparsity constraint ($\|\theta\|_1$). To evaluate the impact of this constraint, we conducted experiments on 10 testing scenes from the KAIST dataset based on the SD-CASSI system. In the experiment, the number of training epochs for the LCTC prior was set to 4,000. Fig. 10 compares the performance of the LCTC prior with and without the sparsity constraint (denoted as ‘no const.’ in the figure) on the network input θ . Metrics including PSNR, SSIM, and SAM were used for evaluation. The results clearly indicate that the sparsity constraint significantly enhances the performance of the LCTC model. For instance, the LCTC method with the ℓ_1 norm

TABLE IV

COMPARISON OF RECONSTRUCTION RESULTS UNDER DIFFERENT INITIALIZATION METHODS FOR LCTC NETWORK WEIGHTS Φ

Metric	Kaiming	Xavier	Gaussian	Uniform	Orthogonal
PSNR	37.67	34.34	35.66	35.31	32.72
SSIM	0.950	0.937	0.942	0.940	0.925
SAM	0.042	0.058	0.048	0.046	0.067

TABLE V

COMPARISON OF RECONSTRUCTION RESULTS UNDER DIFFERENT INITIALIZATION METHODS FOR LCTC NETWORK INPUTS θ

Metric	Ones	Zeros	Gaussian	Uniform	Orthogonal
PSNR	32.85	33.21	37.67	37.12	35.09
SSIM	0.885	0.892	0.950	0.943	0.918
SAM	0.072	0.068	0.042	0.044	0.055

constraint demonstrates a notable improvement over the model without the constraint on Scene 3 (S3). Overall, the inclusion of the sparsity constraint results in an average improvement of 0.79 dB in PSNR, highlighting the effectiveness of enforcing sparsity in optimizing the compressive signal.

3) *Effect of Initialization Method for LCTC:* To investigate the effect of the initialization of network weights and network inputs for LCTC, we conducted a comprehensive ablation study on Scene 1 of the KAIST dataset under SD-CASSI settings, evaluating multiple initialization strategies. Table IV presents the reconstruction results under different initialization methods for the network weights. Among the tested strategies, Kaiming, Xavier, Gaussian, uniform (ranging from -0.1 to 0.1), and orthogonal, Kaiming initialization achieves the best performance across all metrics. Similarly, Table V evaluates different initializations for the network input, including fixed values (ones, zeros), random noise (Gaussian, uniform), and structured inputs (orthogonal). The results show that initializing with Gaussian noise provides a significant advantage.

4) *Ablation Study of the LCTC Network Architecture:* To further assess the effectiveness of the proposed LCTC architecture, we perform an ablation study by replacing the LCTC network with two alternative designs: a 3D CNN-based U-Net and a network equipped with spatial Transformer blocks. The 3D CNN variant adopts a hierarchical encoder-decoder architecture based on volumetric convolutions. It consists of multiple modules with 3D convolutional layers, batch normalization, and ReLU activations, capturing spatial and spectral information simultaneously. Downsampling is performed only in spatial dimensions using 3D max-pooling, while skip connections facilitate spatial detail preservation during decoding. The Transformer-based variant replaces the CTC module with a spatial Transformer block. It consists of a channel-wise self-attention mechanism, followed by a two-layer feed-forward network with GELU activation. Layer normalization is applied before each sub-block.

Both variants follow the same training strategy, loss function, and initialization settings as LCTC to ensure fair comparison. The experiments are conducted under the

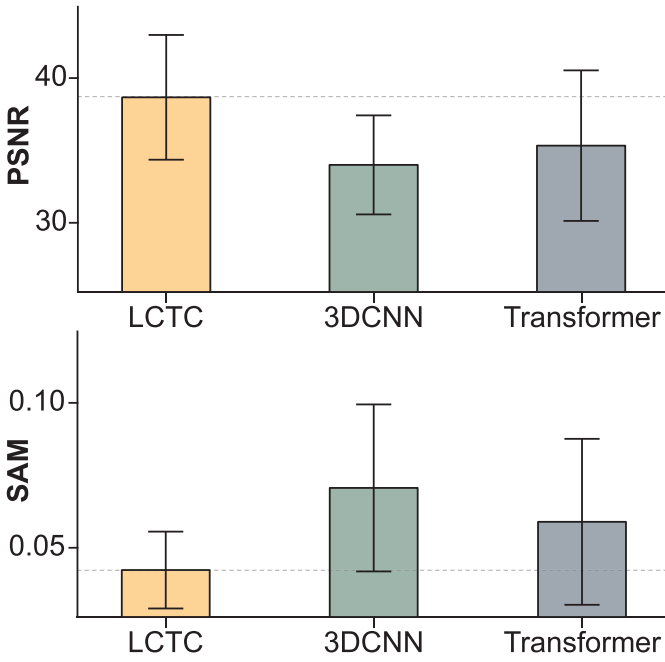


Fig. 11. Ablation comparison of LCTC with 3DCNN and Transformer variants on 10 KAIST scenes.

SD-CASSI configuration using the KAIST dataset. Quantitative results across 10 testing scenes are reported in Fig. 11. As illustrated, the proposed LCTC network consistently surpasses both the 3DCNN and Transformer-based alternatives in terms of PSNR and SAM. These results suggest that although 3D convolutions and Transformer mechanisms are effective in modeling spatial and spectral dependencies, they may be either too generic or too parameter-heavy to generalize well in a self-supervised setting. In contrast, the proposed LCTC design, in a lightweight and specific manner, leads to more robust and accurate reconstruction.

5) *Ablation Study of the CTC Mechanism*: Fig. 12 illustrates the PSNR results of the untrained neural method architecture with the designed Convolutional Thresholding sparse Coding mechanism (denoted as w CTC in the figure) and without Thresholding sparse Coding part (*i.e.*, with only normal convolutional layers and denoted as w/o TC). The experiments were conducted on the 10 scenes from the KAIST dataset based on the SD-CASSI system. The left part of Fig. 12 shows the results of the solo utilization of the LCTC prior, while the right part highlights the impact of CTC within the PnP-LCTC framework. It is evident that the inclusion of the CTC mechanism leads to significant improvements. The mean PSNR values (indicated by dashed lines) of algorithms with CTC are consistently higher than those without the Thresholding Sparse Coding mechanism (w/o TC).

Moreover, we compare the LCTC and PnP-LCTC methods without sparse weights, *i.e.*, removing the soft-shrinkage operation for the thresholding significant weight s_h . We denote it as w/o SW in the figure. The results demonstrate that the sparse operation is crucial for enhancing reconstruction quality, as the dash lines representing mean PSNR values are higher for the methods incorporating the sparse operation compared

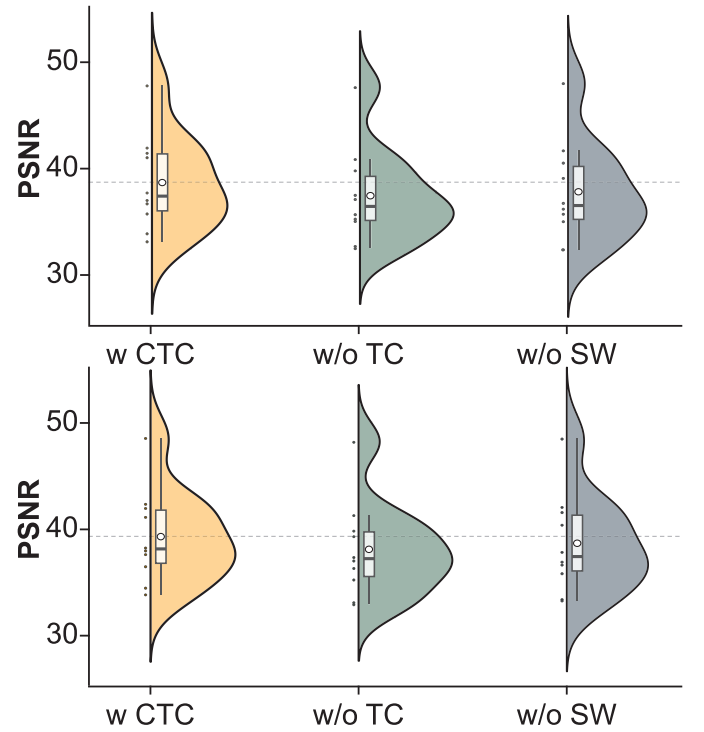


Fig. 12. The effect of the CTC module for the LCTC (top) and PnP-LCTC (bottom) on the 10 testing scenes.

to those without sparse weights (w/o SW). These findings emphasize the importance of the CTC module and sparse operations in boosting the reconstruction performance of the LCTC framework.

6) *Limitation and Further Research*: First, regarding scalability to extremely high spectral resolution data, the optimization time and memory consumption tend to increase with the number of spectral bands, as LCTC is optimized in a per-measurement, self-supervised manner. Although the current design remains efficient for most practical hyperspectral imaging tasks, further improvements, such as dimensionality reduction, may be required to maintain scalability in ultra high-resolution settings. Second, while LCTC is relatively robust to measurement noise, it is more sensitive to degradation in the sensing matrix (*e.g.*, discrepancies between the ideal coded aperture and the physically realized one). In such cases, reconstruction performance may decline. In further work, the degradation-aware algorithm can be developed.

VI. CONCLUSION

This paper aims to reconstruct high-fidelity hyperspectral images from incomplete measurements in Compressive spectral imaging systems. To overcome the limitations of traditional methods, we propose a novel Lightweight untrained neural network (LCTC) prior, a data-driven solution designed to adaptively learn the sparse transformation domain and compressible signal for each captured scene through self-supervised learning. Furthermore, the designed LCTC network, with a compact architecture (only 0.24MB of parameters and 4.7G FLOPs), significantly enhances the ability to identify

the transform domain effectively and efficiently. Additionally, we explore the integration of the LCTC prior into an iterative optimization framework as a plug-and-play prior. Extensive experiments conducted on both simulated and real-world datasets demonstrate that the LCTC prior consistently outperforms state-of-the-art methods in terms of reconstruction quality and computational efficiency. In summary, the proposed LCTC prior offers a robust, efficient, and adaptive solution for hyperspectral image reconstruction.

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